ENSEMBLE MACHINE LEARNING APPROACHES FOR HEART DISEASE RISK PREDICTION

7COM1039 – MSc Project

FEBRUARY 20, 2026

PEREKEME AGBEGHA (24070313)
Paul Moggridge

Contents

1. Background Research and Literature Review 2
1.1 Aim 2
1.2 Research Questions 2
1.3 Objectives 2
1.4 Literature Search Strategy 2
1.5 Inclusion and Exclusion Criteria 3
1.6 Critical Literature Review 3
Paper A: Feature Selection + Ensemble Learning for Heart Disease (2021) .... **Error! Bookmark not defined.**
Paper B: Random Forest with Sequential Feature Selection (2024) ..... **Error! Bookmark not defined.**
Paper C: Supervised ML Prediction on UCI Cleveland (2021)**Error! Bookmark not defined.**
Paper D: Naive Bayes System and Imbalance Effects (2023). **Error! Bookmark not defined.**
1.7 Practical Research and Technical Resolution Examples 4
2. Summary of Progress to Date 5
2.1 Work Completed 5
2.2 Tools, Technologies and Justification 5
2.3 Evidence of Progress (to be included in Appendices) 5
2.4 Challenges and How They Were Handled 5
3. Ethical, Legal, Professional and Social Issues 6
3.1 Ethical Issues ..... **Error! Bookmark not defined.**
3.2 Legal Issues ..... **Error! Bookmark not defined.**
3.3 Professional Issues ..... **Error! Bookmark not defined.**
3.4 Social Issues ..... **Error! Bookmark not defined.**
4. Project Plan 8
4.1 Remaining Tasks and Target Dates 8
4.2 Risk, Quality and Feedback Strategy 9
5. Level of the Project (MSc Justification) 10
References (Harvard) 11
Appendices 12

1. Background Research and Literature Review

Heart disease (cardiovascular disease, CVD) remains a leading global cause of mortality and creates heavy pressure on healthcare systems. Cardiovascular diseases (CVDs) are estimated to cause around 19.8 million deaths each year and account for roughly one third of global deaths. This matters for clinical practice because earlier identification of high-risk patients can support earlier interventions and reduce avoidable harm. However, diagnosis often requires interpreting multiple clinical and demographic factors (e.g., age, blood pressure, cholesterol, chest pain characteristics) that may interact in non-linear ways. Machine learning (ML) can learn patterns from this data and support risk prediction. Yet a single ML model can be unstable (high variance), biased (high bias), or overly sensitive to noise and sampling. Ensemble learning aims to reduce these weaknesses by combining multiple models to obtain more robust and reliable predictions (Dietterich, 2000; World Health Organization, 2025).

1.1 Aim

The aim of this project is to develop and evaluate ensemble machine learning models for predicting heart disease risk using clinical and demographic patient data. The project will compare ensemble approaches against individual machine learning models, focusing on predictive accuracy, robustness, and suitability for healthcare decision-support.

1.2 Research Questions

RQ1: What are the best design choices for building a bagging-based ensemble classifier for heart disease risk prediction (i.e., choice of base learners, how diversity is created via sampling/parameters/features, and how outputs are combined)?

RQ2: What trade-offs exist between predictive accuracy and model interpretability when using ensemble methods in healthcare, and how can explainability techniques mitigate these trade-offs?

1.3 Objectives

- Review relevant literature on heart disease prediction and ensemble machine learning methods.
- Acquire and preprocess a publicly available heart disease dataset.
- Implement baseline (single) models and bagging/ensemble variants.
- Evaluate models using appropriate metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC) and cross-validation.
- Analyse results, including robustness and interpretability, and discuss healthcare implications.

1.4 Literature Search Strategy

A structured literature search was conducted using IEEE Xplore. Search strings included combinations of: “heart disease prediction”, “cardiovascular”, “ensemble”, “bagging”, “voting”, “stacking”, “feature selection”, and “clinical data”. The initial year range was set broadly from 2010–2026 to capture the modern ‘data science’ period while still

allowing inclusion of any influential earlier work when needed. Titles and abstracts were screened first, followed by full-text review of the most relevant papers.

1.5 Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
• Published 2010–2024 • Peer-reviewed journal or IEEE conference • Focuses on heart disease / cardiovascular risk prediction • Uses structured clinical/demographic data (tabular) • Reports quantitative evaluation metrics	• Published before 2010 (unless seminal/required for theory) • Non-peer-reviewed sources (blogs, forums) • Not focused on cardiovascular outcomes • Pure signal-only work without clinical variables (unless directly relevant) • No evaluation metrics / no experimental evidence

1.6 Critical Literature Review

The reviewed literature generally indicates that ensembles can outperform single classifiers on heart disease datasets, but reported gains vary with dataset size, class imbalance handling, feature engineering, and validation design. The review below is grouped by themes to highlight where findings align and where they differ, and to show how these findings motivate the project research questions.

1.6.1 Feature selection and pipeline design

Lakshmanarao et al. (2021) proposed a pipeline combining preprocessing, feature selection (ANOVA and mutual information), imbalance handling (ROS, SMOTE, ADASYN), and multiple classifiers including ensemble voting and stacking. Their results suggest that careful feature selection and resampling can improve performance, but the work also highlights a risk: very high scores after oversampling can reflect leakage or overfitting if validation is not carefully stratified. Gupta et al. (2024) similarly reported strong performance for Random Forest when Sequential Feature Selection was applied. Because these datasets are typically small, near-perfect accuracy claims will be treated cautiously and the project will rely on stratified k-fold cross-validation plus multiple metrics (e.g., ROC-AUC and recall) to avoid misleading conclusions.

1.6.2 Baselines versus ensembles on UCI-style clinical datasets

Hossen et al. (2021) evaluated Logistic Regression, Decision Tree, and Random Forest on the UCI Cleveland dataset and reported competitive performance for Logistic Regression. This is important because it shows that simple, interpretable baselines may be strong on structured tabular data. In healthcare contexts, if a baseline model performs similarly to an ensemble, the simpler model may be preferred due to interpretability and easier clinical accountability. This directly informs RQ2 on the accuracy–interpretability trade-off.

1.6.3 Class imbalance, evaluation choices, and fairness considerations

Sabri et al. (2023) highlighted how class imbalance can distort accuracy in a Naïve Bayes-based heart disease prediction system. They observed that balancing can reduce headline accuracy while improving fairness across classes. This supports prioritising clinically meaningful errors: false negatives (high-risk predicted as low-risk) are often more harmful than false positives in screening-style tasks. Therefore, the project evaluation will prioritise recall, F1-score and ROC-AUC, and will report confusion matrices to make error types visible.

1.6.4 Synthesis: what is agreed, what is unclear, and the gap addressed

Across the reviewed studies, there is broad agreement that (i) robust validation matters as much as model choice, and (ii) preprocessing decisions such as feature selection and imbalance handling can strongly affect results. However, the literature is less consistent on which ensemble 'design choices' are most beneficial for heart disease prediction, because many papers report final scores without systematically testing how base learners are selected, how diversity is created (e.g., bootstrap sampling, feature subsampling, parameter variation), and how predictions are combined. This motivates RQ1, which focuses on ensemble design choices for bagging-based classifiers. In parallel, the evidence that interpretable baselines can be competitive motivates RQ2, which investigates the trade-off between accuracy and interpretability and evaluates explainability methods to support safer decision-support use.

1.7 Practical Research and Technical Resolution Examples

The literature also guided practical design decisions. For example, multiple papers emphasised that clinical datasets can be imbalanced, motivating evaluation using stratified splits and resampling (Lakshmanarao et al., 2021; Sabri et al., 2023). Similarly, feature selection was repeatedly used to reduce dimensionality and improve performance (Lakshmanarao et al., 2021; Gupta et al., 2024). As a result, my implementation prioritises: (i) reproducible preprocessing, (ii) stratified k-fold validation, and (iii) an ensemble design that explicitly creates diversity across base learners via sampling and parameter variation.

2. Summary of Progress to Date

Progress to date has focused on establishing a reliable experimental pipeline and producing evidence of early implementation. The work completed aligns directly with the project objectives: literature review, dataset acquisition and preprocessing, baseline model implementation, and initial evaluation setup.

2.1 Work Completed

- Completed structured literature search on IEEE Xplore and selected core papers aligned with ensemble design and evaluation.
- Selected a publicly available heart disease dataset (UCI Cleveland) and documented its features and target definition.
- Implemented preprocessing steps (encoding, scaling, handling missing values where applicable) and reproducible train/test splits.
- Implemented baseline models: Logistic Regression, Decision Tree, Naive Bayes, Support Vector Machine, and Random Forest.
- Configured evaluation pipeline using stratified k-fold cross-validation, confusion matrix, and ROC-AUC reporting.

2.2 Tools, Technologies and Justification

Python was selected as the implementation language due to its maturity in data science and availability of validated libraries. scikit-learn supports consistent model training, cross-validation, and metric reporting, reducing the risk of implementation errors. pandas and NumPy provide reliable data manipulation. GitHub version control is used to track progress and ensure reproducibility.

2.3 Evidence of Progress (to be included in Appendices)

Appendix 1: Dataset description and feature dictionary (UCI Cleveland).
Appendix 2: Preprocessing pipeline summary (encoding, scaling, split strategy).
Appendix 3: Baseline model training scripts and outputs (accuracy, ROC-AUC).
Appendix 4: Confusion matrices and classification reports for baseline models.
Appendix 5: Git commit log screenshots / repository activity summary.

2.4 Challenges and How They Were Handled

The main challenges encountered so far are (i) small dataset size and potential overfitting, and (ii) selecting evaluation metrics that reflect clinical risk. These have been handled by adopting stratified k-fold cross-validation and reporting multiple metrics. In addition, the project will treat recall for the positive class as a key indicator because false negatives are more harmful in screening-style applications.

3. Ethical, Legal, Professional and Social Issues

This section considers ethical, legal, professional and social issues in the current scope (academic evaluation on a public dataset) and in a plausible future scenario where the model is deployed in a real clinical setting. Because healthcare prediction is high-stakes, these considerations are treated as part of the technical design (e.g., which errors to minimise, what data to collect, and how to explain model outputs).

Ethics approval: This work uses secondary, publicly available and anonymised data (UCI Cleveland) and does not involve recruiting participants or collecting identifiable personal data. Therefore, formal ethics approval is not expected to be required for the current academic evaluation. If the work were extended into a clinical investigation or deployed as a medical device, ethics and governance requirements would likely change and may require formal approvals (Health Research Authority, 2025).

3.1 Ethical Issues

An ethical risk assessment framework (as introduced in Data Mining ethics teaching) is used to structure risks around: harm, fairness, transparency, and accountability. The primary harm risk is incorrect prediction. In screening-style tasks, a false negative (high-risk predicted as low-risk) may delay care, while a false positive may cause anxiety and unnecessary follow-up testing. Because harm from false negatives can be greater, evaluation will prioritise recall and will report confusion matrices and ROC-AUC rather than accuracy alone.

Fairness is another ethical risk: performance can vary across subgroups (e.g., sex or age bands) if training data underrepresents certain groups. Where demographic variables exist in the dataset, subgroup performance checks will be performed (e.g., comparing recall and false negative rates) and findings will be reported transparently. If subgroup gaps are observed, mitigation strategies such as threshold tuning, reweighting, or balanced resampling will be considered and documented.

Transparency and explainability are essential in healthcare because clinicians need reasons they can trust. The project will evaluate post-hoc explainability methods (e.g., SHAP and LIME) to provide feature-level explanations that can be cross-checked against clinical knowledge (Sathyan et al., 2022). Explanations will be treated as decision-support evidence, not proof of causality, to avoid over-trust in model outputs.

3.2 Legal Issues

In the current scope, the dataset is public and anonymised; however, legal duties still apply around responsible handling and correct attribution. UK GDPR principles that are especially relevant include: data minimisation (collect only what is needed), storage limitation (do not retain data longer than necessary), and integrity/confidentiality (appropriate security). In practice, this means storing data securely, avoiding re-identification attempts, and keeping data access limited to the project context (Information Commissioner's Office, 2024).

If the project were deployed in a real NHS environment, additional governance controls would be required, such as compliance with the NHS Data Security and Protection Toolkit (NHS England, 2026). Moreover, software used to influence clinical decisions

may fall under UK medical device regulation and would require an 'intended purpose' statement, safety/performance evidence, and appropriate approvals before use (NHS Digital Regulations, 2024).

3.3 Professional Issues

Professional practice focuses on quality, reproducibility, and accountability. The project will use version control (Git), fixed random seeds, and clear experiment logs to ensure results can be reproduced. Model development will follow a transparent workflow: documentation of preprocessing steps, parameter settings, and evaluation protocols, plus reporting of negative results when ensembles do not outperform baselines.

As a computing professional in the UK context, the work will align with the BCS Code of Conduct, including acting in the public interest, maintaining competence, and being honest about limitations and uncertainty (BCS, 2021). If the system were to be deployed clinically, professional responsibilities would extend to clinical governance, stakeholder engagement, and rigorous validation aligned with NHS and NICE expectations for digital health tools (NICE, 2022).

3.4 Social Issues

Socially, decision-support tools can improve access to early screening and triage by helping clinicians prioritise patients who may need further assessment. In resource-limited settings, this could make risk assessment cheaper and more accessible by using routinely collected clinical attributes rather than expensive tests. However, there is also a risk of widening inequality if models perform worse for underrepresented groups or if digital tools are deployed without considering access, usability, and clinician training.

To support responsible use, performance will be reported transparently (including subgroup checks where possible) and outputs will be framed as decision support rather than automated decision-making. This reduces the risk of automation bias (over-reliance on algorithmic outputs) and supports inclusive, accountable adoption in future work.

4. Project Plan

A hybrid project management approach is adopted, combining a structured timeline with iterative experimentation. Weekly milestones and supervisor feedback will be used to control scope and ensure progress. The plan reserves dedicated time for testing, validation, and final report writing to meet MSc standards.

4.1 Remaining Tasks and Target Dates

Task	Deliverable	Target Date
Implement bagging ensembles (bootstrap sampling + base learners)	Bagging experiment scripts + results table	**20 March 2026**
Design diversity strategies (sampling / feature subsampling / parameter variation)	Diversity ablation study results	**23 March 2026**
Implement aggregation methods (majority voting, weighted voting)	Comparison of combination rules	**25 March 2026**
Hyperparameter optimisation (GridSearchCV/RandomSearchCV)	Best model configs + justification	**28 March 2026**
Interpretability study (baseline vs ensemble + SHAP/LIME)	Explainability examples + discussion	**31 March 2026**
Final evaluation and robustness checks (k-fold, ROC-AUC, calibration)	Final performance and robustness report	**5 April 2026**
Write final dissertation + prepare demo	Complete dissertation and presentation/demo	**27 April 2026**

4.2 Risk, Quality and Feedback Strategy

Key risks include overfitting due to limited dataset size, and misleading conclusions from relying on a single metric. To mitigate this, the project will use stratified cross-validation, multiple metrics, and calibration checks. Quality will be assessed through reproducibility (fixed random seeds and saved configurations), comparative baselines, and clear reporting. Supervisor meetings will be used to review progress, adjust priorities, and confirm that experiments remain aligned with the research questions.

5. Level of the Project (MSc Justification)

This project meets MSc level expectations by combining theory-driven investigation with rigorous experimentation and validation. The work goes beyond applying a single classifier: it systematically studies ensemble design choices (base learner selection, diversity creation, combination rules) and evaluates their impact on robustness and reliability. It also addresses a key healthcare requirement: balancing predictive performance with interpretability and accountability. The planned evaluation includes ablation studies, stratified cross-validation, calibration checks, and explainability analyses, which demonstrate depth, complexity, and technical rigour expected at MSc level.

The project also builds directly on MSc modules (e.g., Data Mining / Machine Learning teaching) by taking taught ensemble concepts and applying them to a realistic healthcare dataset with a stronger emphasis on experimental design, validation, and professional constraints (ethics, GDPR, and clinical accountability).

References (Harvard)

BCS (2021) BCS Code of Conduct. [Online] Available at: https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/ (Accessed: 27 February 2026).

Breiman, L. (2001) 'Random Forests', Machine Learning, 45(1), pp. 5–32. doi:10.1023/A:1010933404324.

Dietterich, T.G. (2000) 'Ensemble Methods in Machine Learning', in Multiple Classifier Systems. Berlin: Springer, pp. 1–15.

Gupta, M., Menon, S.V., Sowmiya, S., Banchhor, C.O., Parmar, Y. and Vigenesh, M. (2024) 'Refining Heart Disease Diagnosis with Machine Learning: Techniques for Optimal Medical Outcomes', 2024 International Conference on Recent Advances in Science & Engineering Technology (ICRASET).

Health Research Authority (2025) Medical devices and software applications. [Online] Available at: https://www.hra.nhs.uk/planning-and-improving-research/policies-standards-legislation/medical-devices-and-software-applications/ (Accessed: 27 February 2026).

Hossen, M.D.A., Tazin, T., Khan, S., Alam, E., Sojib, H.A., Khan, M.M. and Alsufyani, A. (2021) 'Supervised Machine Learning-Based Cardiovascular Disease Analysis and Prediction', Mathematical Problems in Engineering, 2021, Article ID 1792201.

Information Commissioner's Office (2024) Guide to the UK GDPR: Principles. [Online] Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/guide-to-the-uk-gdpr/principles/ (Accessed: 27 February 2026).

Kuncheva, L.I. (2014) Combining Pattern Classifiers: Methods and Algorithms. 2nd edn. Hoboken, NJ: Wiley.

Lakshmanarao, A., Srisaila, A. and Ravi Kiran, T.S. (2021) 'Heart Disease Prediction using Feature Selection and Ensemble Learning Techniques', Proceedings of the Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV 2021).

NHS England (2026) Data Security and Protection Toolkit. [Online] Available at: https://standards.nhs.uk/published-standards/data-security-and-protection-toolkit (Accessed: 27 February 2026).

NHS Digital Regulations (2024) Writing the intended purpose statement for software medical devices. [Online] Available at: https://www.digitalregulations.innovation.nhs.uk/regulations-and-guidance-for-developers/all-developers-guidance/writing-the-intended-purpose-statement-for-software-medical-devices/ (Accessed: 27 February 2026).

NICE (2022) Evidence standards framework for digital health technologies. [Online] Available at: https://www.nice.org.uk/corporate/ecd7 (Accessed: 27 February 2026).

Sabri, N., Shari, A.A., Abu Samah, K.A.F., Mohd Noordin, M.R., Shari, A.S., Ishak@Zainudin, F.M. and Wan Mustapha, W.M. (2023) 'HeartInspect: Heart Disease

Prediction of an Individual Using Naïve Bayes Algorithm', 2023 IEEE 11th Conference on Systems, Process & Control (ICSPC).

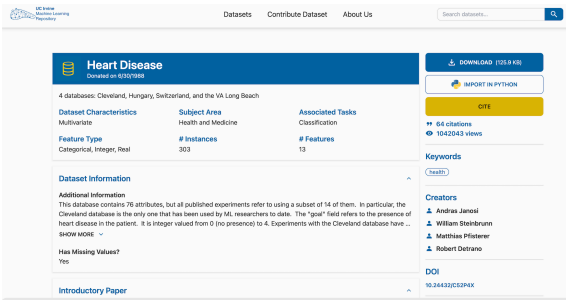
Sathyan, A. et al. (2022) 'Interpretable AI for bio-medical applications', [Online] Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC10074303/ (Accessed: 27 February 2026).

UCI Machine Learning Repository (1988) Heart Disease Data Set (Cleveland). [Online] Available at: https://archive.ics.uci.edu/ml/datasets/Heart+Disease (Accessed: 27 February 2026).

World Health Organization (2025) Cardiovascular diseases. [Online] Available at: https://www.who.int/health-topics/cardiovascular-diseases (Accessed: 27 February 2026).

Appendices

Appendix 1: Dataset source and feature dictionary (UCI Cleveland)



Appendix 2: Preprocessing pipeline screenshots / logs

This appendix provides evidence of the preprocessing pipeline applied to the Cleveland Heart Disease dataset prior to model training. The pipeline includes data inspection, handling of missing values, categorical encoding, target variable transformation, and exploratory data visualisations. These steps ensure data quality, numerical compatibility, and suitability for supervised classification.

```
import pandas as pd

df = pd.read_csv("heart_disease_uci.csv")

display(df.head())

print("Shape:", df.shape)

print(df.isnull().sum())
print(df['num'].value_counts())
```

Figure A2.1: Initial dataset inspection and missing value analysis.

```python
if "id" in df.columns:
    df = df.drop(columns=["id"])

for col in ["fbs", "exang"]:
    if col in df.columns:
        df[col] = df[col].fillna(False).astype(int)

df["num"] = pd.to_numeric(df["num"], errors="coerce")
df["num"] = df["num"].apply(lambda x: 0 if x == 0 else 1)

df = pd.get_dummies(df, drop_first=True)

df = df.fillna(df.median(numeric_only=True))

X = df.drop("num", axis=1)
y = df["num"]

print("Any non-numeric columns left?",
X.select_dtypes(include="object").columns.tolist())
print("Target distribution:\n", y.value_counts())
```

Figure A2.2: Preprocessing pipeline implementation.

Figure A2.3: Class distribution of the binary heart disease target variable.

The class distribution indicates a moderate imbalance, with 509 instances labelled as heart disease (1) and 411 instances labelled as no disease (0). Although not severely imbalanced, this distribution suggests that evaluation metrics beyond simple accuracy (such as recall, F1-score, and ROC-AUC) are necessary to ensure reliable assessment of predictive performance.

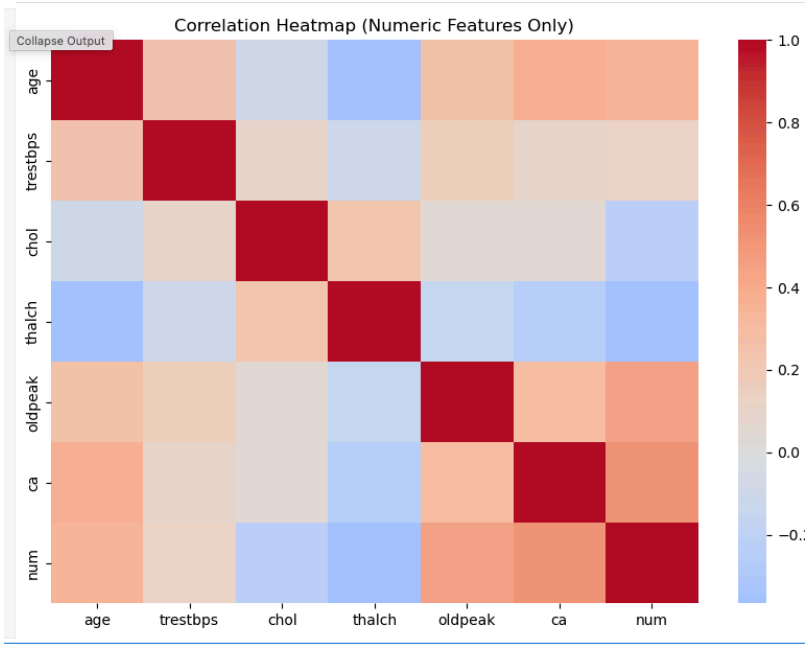


Figure A2.4: Correlation heatmap of numerical features .

The heatmap illustrates the relationships between numerical variables and the binary heart disease outcome. The feature "ca" (number of major vessels) shows the strongest positive correlation with the target variable, followed by "oldpeak" and "age". In contrast, "thalch" (maximum heart rate achieved) exhibits a negative correlation with disease presence. Importantly, no single feature demonstrates extremely high correlation with the outcome, suggesting that heart disease prediction requires modelling interactions among multiple variables rather than relying on a single predictor.

Appendix 3: Baseline model code snippets and outputs

This appendix provides evidence of the baseline classification model implemented for heart disease prediction. A logistic regression classifier was trained using an 80/20 stratified train-test split. Feature scaling was applied using StandardScaler to ensure numerical stability and improve optimisation convergence. Model performance was evaluated using precision, recall, F1-score, and confusion matrix analysis.

```
from sklearn.model_selection import train_test_split

X = df.drop("num", axis=1)
y = df["num"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)



from sklearn.pipeline import make_pipeline
```

```python
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix

model = make_pipeline(
    StandardScaler(),
    LogisticRegression(max_iter=2000)
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
```

Classification report and confusion matrix for baseline logistic regression model.

The baseline logistic regression model achieved an overall accuracy of approximately 83%. The model demonstrates higher recall for the positive (disease) class compared to the negative class, indicating stronger ability to correctly identify patients with heart disease. However, misclassifications are still present, highlighting the potential benefit of ensemble methods for improving predictive robustness.

Appendix 4: Confusion matrices, ROC curves, and metric tables

Appendix 5: Git repository activity log / commit history

Appendix 6: Supervisor meeting notes and action points

Date	Stage	Key Discussion Points	Action Points
6 February 2026	Detailed Project Proposal (DPP)	Discussed project scope and refinement of research questions. Supervisor advised narrowing focus to ensemble design choices rather than broad ML comparison.	Refine research questions to focus on bagging and ensemble construction. Update proposal accordingly.
13 February 2026	DPP Revision	Reviewed updated proposal. Discussed ensemble design elements (base learner choice, sampling strategy, combination methods). Advised adding exploratory data analysis and clearer justification of MSc level.	Add EDA section, clarify ensemble design approach, strengthen MSc-level justification in report.
23 February 2026	Interim Progress Report (IPR)	Reviewed implementation progress. Supervisor emphasised need for visual exploratory analysis (class imbalance, correlations) and clear evidence of baseline model implementation.	Add class distribution plot, correlation heatmap, baseline model outputs, and organise appendices clearly.